Tolman-Oppenheimer-Volkoff Equation

The general relativistic form of the TOV equation is as follows [1].

$$\frac{dP(r)}{dr} = -\frac{Gm_r(r)\varepsilon(r)}{r^2}\left(1+\frac{P(r)}{\varepsilon(r)}\right)\left(1+\frac{4\pi r^3 P(r)}{m_r(r)}\right)\left(1-\frac{2Gm_r(r)}{r}\right)^{-1}$$

r = Radius

P = Pressure

ε = Energy Density

m_r = Enclosed Mass

G = Gravitational Constant

Dimensionless TOV Equation

By introducing the following scaling constants, it's possible to convert the TOV equation to a dimensionless form.

$$P = \varepsilon_0 \cdot P'$$
$$\varepsilon = \varepsilon_0 \cdot \varepsilon'$$
$$r = a \cdot r'$$
$$m_r = b \cdot m_r'$$

All of the primed variables represent dimensionless quantities of the corresponding value. ε_0 has units of energy density/pressure, a has units of length, and b has units of mass.

After plugging into the TOV equation, there are still some dimensional quantities left. To cancel those, the following equations must be true.

$$G \cdot b = a$$
$$a^3 \cdot \varepsilon_0 = b$$

These equations hold true for

$$a = (G \cdot \varepsilon_0)^{-1/2}$$
$$b = (G^3 \cdot \varepsilon_0)^{-1/2}$$

The fully dimensionless TOV equation reads

$$\frac{dP'}{dr'} = -\frac{m'_r \varepsilon'}{r'^2}\left(1 + \frac{P'}{\varepsilon'}\right)\left(1 + \frac{4\pi r'^3 P'}{m'_r}\right)\left(1 - \frac{2m'_r}{r'}\right)^{-1}$$

The mass-conservation equation scales as one would expect

$$\frac{dm'_r}{dr'} = 4\pi r'^2 \varepsilon'$$

Note that the equations are now reliant on an EoS of the form $P'(\varepsilon')$.

Sources

1. Compact Star Physics by Jürgen Schaffner-Bielich